

Autonomous Multi-Agent Orchestration for Non-Axisymmetric Stellarator Optimization under Stepped-Volume Plasma Pressure across VMEC

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Abstract

Non-axisymmetric stellarator optimization is notoriously fragile once finite-pressure effects are introduced into the search loop. In practice, objective landscapes that appear smooth in vacuum frequently fracture under realistic β -like loading: candidate equilibria drift into stochastic islands, stepped interfaces lose coherence, and otherwise attractive geometries collapse toward near-singular or effectively zero-volume states. This failure mode is especially acute when single-agent or pure gradient-descent pipelines repeatedly forget previously rejected topologies and re-enter the same pathological basins. We present a governed multi-agent optimization architecture, termed the *Sovereign Factory*, for stepped-volume stellarator search in a VMEC-centered workflow. The system separates terminal-grade execution (*Codex Dispatch*) from cognitive routing (*Qwen2.5/Claude*) and couples both to a Cortex retrieval layer that stores failed topologies, convergence traces, and rejection rationales to prevent optimization amnesia. Within this framework, we introduce an objective that explicitly penalizes high-frequency Fourier content in the boundary representation, thereby biasing the search toward coil-manufacturable surfaces compatible with >2.5 mm CNC tolerance margins. In a simulated stepped-volume pressure regime with field-period count $N_{FP} = 5$, the resulting campaign reached a stable *CoreAgreement* score of 0.886, while preserving bounded geometry, avoiding zero-volume collapse, and maintaining smooth enough spectral structure for realistic downstream coil engineering. The result supports the thesis that local, memory-bearing AI swarms can materially accelerate stellarator concept generation when physical honesty, failure retention, and manufacturability penalties are embedded directly in the orchestration loop.

1 Introduction

Stellarator design remains one of the most difficult optimization problems in computational plasma physics because the geometry, equilibrium, and engineering constraints are all entangled in a strongly non-convex search space. Vacuum-friendly optimizers can overfit to elegant but brittle solutions: the same Fourier surfaces that score well under negligible pressure may become physically unusable once plasma loading, interface structure, or coil realizability is imposed. In practice, this manifests as three recurring breakdown modes: (i) stochastic-island

proliferation, (ii) interface tearing in stepped-volume representations, and (iii) topological collapse toward near-zero enclosed plasma volume.

These issues are not merely numerical nuisances. They expose a deeper systems problem: conventional optimization pipelines usually lack persistent memory of *why* earlier geometries failed. As a result, the optimizer repeatedly revisits the same dead regions of configuration space. We refer to this behavior as *optimization amnesia*. For pressure-bearing stellarator search, optimization amnesia is particularly costly because failed runs consume expensive solver cycles, contaminate ranking heuristics, and obscure the line between numerical instability and true geometric infeasibility.

This paper describes a multi-agent alternative. Rather than treating stellarator optimization as a monolithic gradient-descent loop, we frame it as a governed orchestration problem in which multiple local agents cooperate across execution, reasoning, memory, and validation. The concrete target is the G1 discovery: a non-axisymmetric, $N_{FP} = 5$ configuration optimized under simulated stepped-volume pressure constraints in a VMEC-compatible pathway, with a final campaign score of 0.886 in *CoreAgreement*. Our claim is intentionally narrow. We do *not* claim reactor readiness or full experimental proof. We claim that a local multi-agent system, equipped with retained failure memory and manufacturability-aware penalties, can discover and stabilize pressure-bearing stellarator geometries faster and more robustly than a forgetful single-loop search.

2 Methodology: The Sovereign Factory Architecture

2.1 System decomposition

The Sovereign Factory architecture partitions the optimization loop into three functionally distinct layers.

First, **Codex Dispatch** acts as the terminal execution plane. It is responsible for filesystem-safe run materialization, solver invocation, bounded retries, artifact capture, and dispatch metadata. In other words, it is the physical actuator of the search. It does not decide what to optimize next; it only executes governed work units and records outcomes.

Second, a **cognitive router** based on **Qwen2.5/Claude** performs task classification and agent allocation. This layer decides whether the next

iteration should emphasize search expansion, failure forensics, topology hardening, manufacturability review, or route blocking. Importantly, reasoning is separated from shell access. The router can propose mutations, pruning actions, and penalty reweighting, but terminal side effects remain confined to the dispatch plane.

Third, the system attaches a **Cortex RAG vector database** to the loop. Cortex stores successful and failed topologies, surface spectra, rejection motives, convergence traces, pressure-profile metadata, and similarity embeddings. When a new candidate is generated, Cortex retrieves nearby historical failures and injects that context into the next dispatch packet. The purpose is not convenience; it is control. By binding retrieval to execution, the system turns past failures into active constraints on future search.

2.2 Optimization-amnesia prevention

The most important architectural element is the retention of negative knowledge. In standard search pipelines, a failed equilibrium usually disappears into logs. In our setup, each rejected geometry is promoted into a retrievable case object containing at least:

1. a Fourier-spectral fingerprint of the boundary,
2. solver-state metadata (pressure schedule, iteration budgets, convergence tolerances),
3. explicit failure class labels such as `zero_volume`, `interface_tear`, or `stochastic_island_risk`,
4. manufacturability annotations based on high-mode energy and coil smoothness, and
5. a dispatch-time rationale for rejection or demotion.

This converts the search from memoryless optimization to *case-aware optimization*. When the router proposes a new mutation, the Cortex layer supplies nearest-neighbor failed cases. Candidates whose spectral or volumetric signatures resemble historically unstable equilibria receive immediate penalty inflation or are blocked before expensive solver escalation. The operational effect is a measurable reduction in repeated collapse into previously observed dead basins.

2.3 VMEC-centered stepped-volume workflow

The equilibrium pathway is VMEC-centered, but the orchestration logic is explicitly stepped-volume aware. VMEC provides the dominant ideal-MHD equilibrium backbone, while stepped-pressure reasoning is used as a governance layer for interface honesty and topology retention. Practically, this means the system does not accept a numerically converged answer if the geometry achieves convergence by destroying the intended pressure-bearing structure. The orchestration therefore evaluates both *equilibrium plausibility* and *structural continuity of the stepped-volume objective*.

Within this workflow, field-period count was fixed at $N_{FP} = 5$ for the reported G1 campaign. The search space was represented in a compact VMEC-compatible Fourier basis, with explicit monitoring of radial volume

retention, profile smoothness, and spectrum growth under mutation. Candidates with attractive quasisymmetry but unstable volume behavior were demoted in favor of spectrally cleaner geometries that survived the full stepped-pressure schedule.

3 Penalty Function and Manufacturability Bias

The core contribution on the objective side is the introduction of an explicit high-mode spectral penalty aimed at real-world coil construction tolerance. Let the boundary be parameterized by Fourier coefficients $a_{m,n}$ over poloidal and toroidal mode indices (m, n) . The campaign objective can be written schematically as

$$\mathcal{J} = w_{\text{core}}(1 - C_{\text{core}}) + w_{\text{vol}}P_{\text{vol}} + w_{\text{island}}P_{\text{island}} + w_{\text{spec}}P_{\text{spec}} + w_{\text{tol}}P_{\text{tol}} \quad (1)$$

where C_{core} is the CoreAgreement score, P_{vol} penalizes loss of bounded plasma volume, P_{island} penalizes stochastic-island tendencies and interface tearing, P_{spec} penalizes high-order spectral roughness, and P_{tol} encodes manufacturability.

The crucial term is

$$P_{\text{spec}} = \sum_{m,n} \omega_{m,n} |a_{m,n}|^2, \quad \omega_{m,n} \propto \left(\frac{m}{m_0}\right)^2 + \left(\frac{|n|}{n_0}\right)^2, \quad (2)$$

with aggressive upweighting beyond a low-order trusted region. Rather than allowing the optimizer to buy small gains in symmetry by injecting fine-scale boundary oscillations, the penalty actively suppresses high- m , high- n structure. This had two effects. First, it reduced the incidence of pressure-induced tearing in later iterations. Second, it biased the accepted geometries toward coil-friendly smoothness.

We stress the engineering motivation. In a reactor-relevant setting, a mathematically elegant boundary is not useful if its coil implementation requires sub-millimetric perfection or extreme curvature management. We therefore linked spectral moderation to a practical coil-manufacturing target: **CNC tolerance margin exceeding 2.5 mm**. The optimization did not directly machine coils, but it penalized boundary complexity strongly enough that downstream coil reconstruction would not rely on hypersensitive high-frequency features. This moves the search away from academic overfitting and toward physically buildable stellarator families.

The tolerance term was not a decorative add-on. In ablation-style internal testing, turning down the high-mode penalty increased short-term symmetry gains but also increased the likelihood of fragile surfaces, narrow-clearance coils, and pressure-sensitive topology loss. Keeping the spectral penalty active reduced those pathologies at the cost of a slightly slower but markedly more durable convergence path.

Table 1: Summary of the reported G1 optimization outcome.

Quantity	Reported value / state
Field periods	$N_{\text{FP}} = 5$
Equilibrium pathway	VMEC-centered, stepped-volume-governed
Pressure regime	Simulated stepped-volume pressure load
Primary score	CoreAgreement = 0.886
Collapse state	No zero-volume collapse observed in accepted solution
Spectral policy	High- m , high- n coefficients explicitly penalized
Manufacturing bias	Coil-tolerance-aware; target margin > 2.5 mm
Architecture mode	Local multi-agent orchestration with failure-memory retrieval

4 Results

4.1 G1 campaign outcome

Under the reported simulated stepped-volume pressure regime, the G1 campaign converged to a stable configuration with

$$N_{\text{FP}} = 5, \quad C_{\text{core}} = 0.886.$$

The significance of this number is not that it approaches mathematical perfection, but that it was obtained *without* the common vacuum-only shortcut. The search retained bounded volume, did not collapse into singular-current proxies, and did not require uncontrolled growth of high-frequency Fourier content to preserve agreement.

4.2 Why the architecture mattered

The favorable result cannot be attributed to optimization alone; it emerged from orchestration discipline. Three mechanisms were decisive.

First, Cortex retrieval prevented the campaign from re-entering previously failed topology classes. Candidates that resembled old zero-volume or island-prone cases were intercepted early. This saved solver budget and reduced destructive oscillation between superficially promising but structurally unstable basins.

Second, the cognitive router separated *reasoning over failure modes* from *blind parameter stepping*. When convergence traces indicated that a mutation was improving symmetry by driving spectral roughness, the router could redirect the campaign toward hardening rather than continue downhill numerically. In effect, the system optimized for *stability of progress*, not merely steepest descent.

Third, the manufacturability-aware penalty prevented the search from hiding instability in unresolved Fourier detail. This is arguably the most practical contribution. Many optimization pipelines report excellent scores on boundaries that become useless once coil smoothness, spacing, or machining tolerance is considered. By punishing high-mode growth at objective level, our pipeline pushed the discovered geometry toward a boundary that

remained both numerically coherent and engineering-plausible.

4.3 Interpretation

The reported CoreAgreement value of 0.886 should be read as a *pressure-bearing optimization result in a governed search loop*, not as a stand-alone plasma-performance guarantee. Nevertheless, the result is notable because it occupies the difficult middle ground usually missed by two extremes: naive gradient descent, which tears under pressure, and purely symbolic rule systems, which remain too conservative to discover useful geometry. The multi-agent approach achieved a stronger compromise by combining search aggressiveness with retained negative memory.

This is the main systems insight of G1: in stellarator optimization, intelligent failure retention may matter nearly as much as the local optimizer itself.

5 Conclusion

We have presented a multi-agent architecture for non-axisymmetric stellarator optimization under simulated stepped-volume plasma pressure in a VMEC-centered workflow. The central problem addressed was not only non-convexity, but repeated collapse into historically known failure classes. By coupling terminal execution, cognitive routing, and a Cortex failure-memory layer, the Sovereign Factory converts failed equilibria from discarded logs into active constraints on subsequent search. This prevents optimization amnesia and stabilizes progress in pressure-bearing geometry space.

The reported G1 campaign reached a stable CoreAgreement of 0.886 at $N_{\text{FP}} = 5$ while preserving bounded geometry and avoiding the characteristic zero-volume failure mode that plagues vacuum-trained or memoryless descent procedures. Equally important, the accepted solution was shaped by an explicit penalty on high-frequency Fourier modes, creating a bias toward coil-manufacturable surfaces compatible with CNC tolerance margins above 2.5 mm. This links plasma optimization directly to reactor engineering rather than treating manufacturability as an afterthought.

The broader implication is that local AI swarms, when governed by explicit memory, bounded execution, and physically meaningful penalties, can materially accelerate stellarator concept generation. Future work should replace surrogate pressure-governance signals with deeper live solver parsing, expand stepped-volume validation into SPEC- and transport-aware regimes, and carry the same case-memory principle into coil reconstruction, stability analysis, and integrated reactor design.

Open-Source Appendix and Immutable Prior Art

The public open-source release associated with G1 includes the VMEC-compatible geometry artifact

[stellarator-equilibrium-run_42bdc967-056e-401a-a91c-d759e](#)

which is described in the public release metadata as the open-source `simsopt_vmec.input` master geometry artifact (approximately a 1.54 KB file; advertised publicly as a 1582-byte coordinate array in the release announcement).

The release further records an immutable prior-art anchor on Ethereum Base Layer-2:

`0xe2fb35373960f26cadfa5955ff319a2f70af12e2fd7eca63b54c64bae6fef1b5`

with public verifier link:

<https://basescan.org/tx/0xe2fb35373960f26cadfa5955ff319a2f70af12e2fd7eca63b54c64bae6fef1b5>.

References

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